Online appendix for “Convergence-divergence models: Generalized phylogenetic trees modeling gene flow over time” for peer review

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Sections, algorithms, definitions, theorems, propositions, figures and expressions introduced in the main text are labeled numerically. Any introduced in the online appendix are labeled with a number followed by “A”.

1A Parameter identifiability

Recall that we decompose edges of the principal tree into “diverging sections” and “converging sections”. Converging sections span only a single epoch, while diverging sections may span multiple epochs.

Recall that rates and epoch times cannot be identified individually; only their products can be identified. A parameter associated with a diverging section of an edge than spans a single epoch cannot be identified. Instead, an "average" over the maximum number of epochs a contiguous diverging section can span can be identified. For example, suppose rate matrix $\boldsymbol{Q}_1$ applies over epoch time t_1 to a diverging section of an edge immediately before an event and rate matrix $\boldsymbol{Q}_2$ applies over epoch time t_2 to a diverging section of the edge immediately after the event. Then for the 2-state general Markov model, $\exp\left(\widehat{\boldsymbol{Q}}\,(t_1+t_2)\right) = \exp\left(\boldsymbol{Q}_2 t_2\right)\exp\left(\boldsymbol{Q}_1 t_1\right)$, where $\widehat{\boldsymbol{Q}}$ is again a rate matrix from the 2-state general Markov model. Thus, we apply rate matrix $\widehat{\boldsymbol{Q}}$ to both diverging sections of the edge.

This lack of identifiability result follows from the 2-state general Markov model forming a Lie algebra, sufficient for multiplicative closure of the model class (Sumner et al. 2012a). Suppose α_1 and β_1 and α_2 and β_2 correspond with rate matrices $\boldsymbol{Q}_1$ and $\boldsymbol{Q}_2$, respectively. Then by Definition 4, $\frac{\alpha_1}{\beta_1} = \frac{\alpha_2}{\beta_2}$. It is straightforward to show that if $\widehat{\alpha}$ and $\widehat{\beta}$ are associated with $\widehat{\boldsymbol{Q}}$, then $\frac{\widehat{\alpha}}{\widehat{\beta}} = \frac{\alpha_1}{\beta_1} = \frac{\alpha_2}{\beta_2}$. Thus, the product of the two transition matrices is replaced by a single "average" transition matrix.

All parameters except for those corresponding to the root distribution are of the form $l_i = a_i + b_i = \alpha_i t_i + \beta_i t_i = a_i\left(1 + \frac{b_j}{a_j}\right)$, where i and j are arbitrary parameter indices, $a_i = \alpha_i t_i$ and $b_i = \beta_i t_i$. (Note that these parameters are scalars, whereas $\exp\left(\widehat{\boldsymbol{Q}}\,(t_1+t_2)\right)$ is a matrix.) Since $\frac{\alpha_i}{\beta_i} = \frac{\alpha_j}{\beta_j}$, it follows that $\frac{a_i}{b_i} = \frac{a_j}{b_j}$.

Contiguous diverging sections of an edge — not separated by a converging section — each have a single associated parameter l_i. Furthermore, each convergence group has an associated parameter l_j, in common for all converging sections of edges in the convergence group. In addition to parameters describing the convergence groups and contiguous diverging sections, there is a parameter $\gamma = [\boldsymbol{\Pi}]_0 - [\boldsymbol{\Pi}]_1 = \frac{-a_i+b_i}{a_i+b_i}$ describing the difference in probabilities of states 0 and 1 on the root taxon.

To form the set of parameters of a CDM, we consider a particular unique set of diverging and converging sections. Since differences in parameters between contiguous diverging sections cannot be identified, the diverging sections we consider are those sections on the principal tree between a node or converging section and another node or converging section. Furthermore, since the exact root location on the outgroup edge is not identifiable, we consider one diverging section to be the entire outgroup edge when the principal tree of the CDM is unrooted. The converging sections correspond to individual epochs where there is a convergence group. Converging sections correspond to convergence parameters and diverging sections correspond to divergence parameters.

Note that although this is the general formulation of the parameter space, on a given CDM not all parameters are necessarily identifiable. To obtain an identifiable set of parameters some combinations of the divergence parameters may be required, which we describe in Section 2A.1. For the following sections, the parameters $x_i = \exp(-l_i) \in (0, 1)$ are used for establishing identifiability and distinguishability of CDMs.

2A Identifiability of 4-taxon CDMs

Here we establish whether each 4-taxon CDM is identifiable or not. A 4-taxon CDM is identifiable if there is a one-to-one mapping from the set of generic parameters to the set of realizable phylogenetic tensors (transformed into the Hadamard basis for simplicity). We start by describing the form of the phylogenetic tensors, then proving that, given an arbitrary phylogenetic tensor, a unique parameter set as functions of the phylogenetic tensor exists.

Sumner et al. (2012b) formally describe phylogenetic epoch models in their Definition 6.1 and introduce notation to compute the phylogenetic tensors. We use the same notation for our CDMs.

For each 4-taxon CDM, the phylogenetic tensor $\boldsymbol{P}$ is transformed into the Hadamard basis $\widehat{\boldsymbol{P}}$ by multiplying by $\boldsymbol{H}_{16} = \boldsymbol{H}_2^{\otimes 4}$, where

$$\boldsymbol{H}_2 = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

In the Hadamard basis, the phylogenetic tensor for CDM 5 is

$$
\widehat{\boldsymbol{P}} = \boldsymbol{H}_{16} \cdot \boldsymbol{P} = \begin{bmatrix} q_{0000} \\ q_{0001} \\ q_{0010} \\ q_{0011} \\ q_{0100} \\ q_{0101} \\ q_{0110} \\ q_{0111} \\ q_{1000} \\ q_{1001} \\ q_{1010} \\ q_{1011} \\ q_{1100} \\ q_{1101} \\ q_{1110} \\ q_{1111} \end{bmatrix}
= \begin{bmatrix} 1 \\ \gamma \\ \gamma \\ \gamma^2 + \left(1-\gamma^2\right) r_{0011} \\ \gamma \\ \gamma^2 + \left(1-\gamma^2\right) r_{0101} \\ \gamma^2 + \left(1-\gamma^2\right) r_{0110} \\ \gamma\left(\gamma^2 + \left(1-\gamma^2\right)\left(r_{0011} + r_{0101} + r_{0110} - 2r_{0111}\right)\right) \\ \gamma \\ \gamma^2 + \left(1-\gamma^2\right) r_{1001} \\ \gamma^2 + \left(1-\gamma^2\right) r_{1010} \\ \gamma\left(\gamma^2 + \left(1-\gamma^2\right)\left(r_{0011} + r_{1001} + r_{1010} - 2r_{1011}\right)\right) \\ \gamma^2 + \left(1-\gamma^2\right) r_{1100} \\ \gamma\left(\gamma^2 + \left(1-\gamma^2\right)\left(r_{0101} + r_{1001} + r_{1100} - 2r_{1101}\right)\right) \\ \gamma\left(\gamma^2 + \left(1-\gamma^2\right)\left(r_{0110} + r_{1010} + r_{1100} - 2r_{1110}\right)\right) \\ \gamma^2\left(\gamma^2 + \left(1-\gamma^2\right)\left(r_{0011} + r_{0101} + r_{0110} + r_{1001} + r_{1010} + r_{1100} \right.\right. \\ \left.\left. -2\left(r_{0111} + r_{1011} + r_{1101} + r_{1110} - 2\delta\right)\right)\right) + \left(1-\gamma^2\right)^2 r_{1111} \end{bmatrix}. \tag{1A}
$$

We see immediately that $\gamma = q_{0001} = q_{0010} = q_{0100} = q_{1000}$. Thus, we can express r_{0011}, r_{0101}, …, r_{1111} and δ as functions of the phylogenetic tensor elements and γ. Finally, we can express the parameters of the CDMs as functions of r_{0011}, r_{0101}, …, r_{1111} and δ. For this final step, we use algebraic geometry, including ideals and their Gröbner bases.

See Mathematica file S2.nb (text version S3.txt) on https://github.com/jonathanmitchell88/CDMsSI for a derivation of Equation (1A) and equations for r_{0011}, r_{0101}, …, r_{1111} and δ in terms of x_i and y_i (products of x_i) for CDM 5. CDMs $1-4$ are all nested in CDM 5. Thus, their phylogenetic tensors are also in the form of Equation (1A). However, the equations for $r_{0011}, r_{0101}, \ldots, r_{1111}$ and δ involve different expressions of x_i and y_i.

For the proof that follows, the order of parameters is as in Figure 2, with $x_i = \exp\left(-\left(a_i + b_i\right)\right) \in (0,1)$ for all $i \in \{1, 2, \ldots, 11\}$. Note again that the exact location of the root on the outgroup edge is not identifiable; t_1 corresponds to the sum of epoch times of epochs from the root to the outgroup added to the first epoch time below the root.

To establish whether a CDM is identifiable or not, we must first determine a maximal set of independent elements of the transformed phylogenetic tensor. That is, a set with maximum cardinality such that there are no algebraic equations — equalities — involving multiple elements of the set. If the cardinality of the set equals the number of parameters, then the CDM is identifiable. For example, we can see that invariants $q_{0001} = q_{0010} = q_{0100} = q_{1000} = \gamma$ are equalities on all CDMs. Thus, we can only include one of q_{0001}, q_{0010}, q_{0100} and q_{1000} in the set.

To determine all such equalities, for a given CDM with $l + 1$ parameters $x_1, x_2, \ldots, x_l, \gamma$, we construct the ideal,

$$I = \langle r_{0011} - f_{0011}\left(x_1, x_2, \ldots, x_l\right), r_{0101} - f_{0101}\left(x_1, x_2, \ldots, x_l\right), \ldots,$$

$$
\begin{aligned}
& r_{1111} - f_{1111}\left(x_1, x_2, \ldots, x_l\right), \delta - f_\delta\left(x_1, x_2, \ldots, x_l\right)\rangle \\
& \subseteq \mathbb{Q}\left[x_1, x_2, \ldots, x_l, r_{0011}, r_{0101}, \ldots, r_{1111}, \delta\right],
\end{aligned}
$$

where each $r_{ijkl} - f_{ijkl}\left(x_1, x_2, \ldots, x_l\right)$ and $\delta - f_\delta\left(x_1, x_2, \ldots, x_l\right)$ is identically zero. (We can ignore γ since $q_{ijkl} = \gamma^2 + \left(1 - \gamma^2\right) r_{ijkl}$ and including any of these invariants does not help us to determine any invariants involving multiple variables r_{0011}, r_{0101}, $\ldots$, r_{1111}, δ.)

In the Macaulay2 file S4.m2 (output file S5.txt) on https://github.com/jonathanmitchell88/CDMsSI we derive the (reduced) Gröbner basis for this ideal for a particular monomial order for CDM 5. Below we outline how this Gröbner basis is computed.

In the Mathematica file S2.nb (text version S3.txt) we derive the following equations to input into the generators of the ideal:

$$\begin{cases} f_{0011} = x_4x_5x_6x_7x_9x_{10}, \\ f_{0101} = x_{10}x_{11}\left(1 - x_9\left(1 - x_2x_3x_4x_6x_8\right)\right), \\ f_{0110} = x_7x_8x_9x_{11}\left(1 - x_6\left(1 - x_2x_3x_5\right)\right), \\ f_{0111} = x_2x_3x_4x_5x_6x_7x_8x_9x_{10}x_{11}, \\ f_{1001} = x_1x_2x_4x_9x_{10}, \\ f_{1010} = x_1x_2x_5x_6x_7, \\ f_{1011} = x_1x_2x_4x_5x_6x_7x_9x_{10}, \\ f_{1100} = x_1x_3x_6x_8x_9x_{11}, \\ f_{1101} = x_1x_2x_3x_4x_6x_8x_9x_{10}x_{11}, \\ f_{1110} = x_1x_2x_3x_5x_6x_7x_8x_9x_{11}, \\ f_{1111} = x_1x_7x_{10}x_{11}\left(x_4x_8x_9\left(x_2\left(1 - x_6\right) + x_3x_5x_6\right) + x_2x_5x_6\left(1 - x_9\right)\right), \\ \quad f_\delta = x_1x_2x_3x_4x_5x_6x_7x_8x_9x_{10}x_{11}. \end{cases} \tag{2A}$$

The functions $f_{0011} = f_{0011}(x_1, x_2, \ldots, x_l)$, $f_{0101} = f_{0101}(x_1, x_2, \ldots, x_l)$, $\ldots$, $f_{1111} = f_{1111}(x_1, x_2, \ldots, x_l)$ and $f_\delta = f_\delta(x_1, x_2, \ldots, x_l)$ depend on the CDM in question, for example, CDM 5.

The monomial order is the elimination order, eliminating the block $x_1, x_2, \ldots, x_l$, with graded reverse lexicographic order on each block, $x_1 > x_2 > \ldots > x_l$ and $r_{0011} > r_{0101} > \ldots > r_{1111} > \delta$.

Next, we compute the (reduced) Gröbner basis I_G of I. Then $I_{G,q} = I_G \cap \mathbb{R}[r_{0011}, r_{0101}, \ldots, r_{1111}, \delta]$ is a Gröbner basis for the elimination ideal involving only $r_{0011}, r_{0101}, \ldots, r_{1111}, \delta$.

Note that q_{1111} is a function of both r_{1111} and δ, the only element of $\widehat{\boldsymbol{P}}$ that is a function of either. Thus, the maximum cardinality set can include at most one of r_{1111} and δ. In S4.m2 we find that when eliminating r_{1111} there are no generators that involve δ. Thus, r_{1111} is eliminated and δ is another independent variable of the system when r_{1111} is eliminated.

Note that there are still some algebraic equations — equalities — involving some elements of $\{r_{0011}, r_{0101}, \ldots, r_{1110}, \delta\}$. In S4.m2 (output file S5.txt) we find the largest cardinality subset with no algebraic equations involving multiple elements. This cardinality, plus one for γ, is the degrees of freedom of the phylogenetic tensor. Given a set of parameters of the CDM, if this degrees of freedom is less than the number of parameters, then the system of polynomial equations is underdetermined and that set of parameters is not identifiable. (Note that some individual parameters may still be indentifiable.) Otherwise, the set of parameters is identifiable. If that set of parameters is not identifiable, it may be possible to combine the parameters in a such a way that the new set of parameters is identifiable.

2A.1 Proof of Proposition 1

See S4.m2 (output file S5.txt) and S6.m2 (output file S7.txt) on https://github.com/jonathanmitchell88/CDMsSI for the computations of the (reduced) Gröbner bases of the ideals in this proof.

Proposition 1 *CDM 5, with parameter set $\{\gamma, y_1, y_2, \ldots, y_9\}$, is identifiable.*

Proof In S4.m2 (output file S5.txt), we see that there are 9 elements of $\{r_{0011}, r_{0101}, \ldots, r_{1110}, \delta\}$ that are free to vary. However, CDM 5 has 11 parameters excluding γ. Thus, this set of parameters is not identifiable. However, recall in Section 1A that taking some products of x_i parameters may be required to obtain a set of identifiable parameters. Since there are 9 elements of $\{r_{0011}, r_{0101}, \ldots, r_{1110}, \delta\}$ that are free to vary, we desire a set of 9 parameters.

In S2.nb (text version S3.txt), we express $f_{0011}, f_{0101}, \ldots, f_{1111}, \delta$ in terms of the set of parameters $\{y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8, y_9\}$. Precisely,

$$\begin{cases} y_1 = x_1, \\ y_2 = x_2, \\ y_3 = x_3 x_8 x_{11}, \\ y_4 = x_4 x_{10}, \\ y_5 = x_5 x_7, \\ y_6 = x_6, \\ y_7 = x_7 x_8 x_{11}, \\ y_8 = x_9, \\ y_9 = x_{10} x_{11}. \end{cases}$$

In S6.m2 (output file S7.txt), we see that this set of parameters is identifiable. We note that $x_i \in (0,1)$ for all $i \in \{1, 2, \ldots, 11\}$. It follows that $r_{0011}, r_{0101}, \ldots, r_{1111}, \delta \in (0,1)$ and $y_i \in (0,1)$ for all $i \in \{1, 2, \ldots, 9\}$. In S2.nb (text version S3.txt), we see that the solutions to the system are

$$
\begin{cases}
y_1 = \frac{\delta}{r_{0111}}, \\
y_2 = \frac{r_{0111}\sqrt{r_{1001}r_{1010}}}{\delta\sqrt{r_{0011}}}, \\
y_3 = \frac{\delta}{\sqrt{r_{0011}r_{1001}r_{1010}}}, \\
y_4 = \frac{r_{1101}\delta\sqrt{r_{0011}}}{r_{0111}r_{1100}\sqrt{r_{1001}r_{1010}}}, \\
y_5 = \frac{\delta}{r_{1101}}, \\
y_6 = \frac{r_{1101}\sqrt{r_{0011}r_{1010}}}{\delta\sqrt{r_{1001}}}, \\
y_7 = \frac{\delta\left(r_{0110}r_{1101}\delta\sqrt{r_{0011}}-r_{0111}^2 r_{1100}\sqrt{r_{1001}r_{1010}}\right)}{r_{0111}r_{1100}\sqrt{r_{0011}r_{1001}}(\delta\sqrt{r_{1001}}-r_{1101}\sqrt{r_{0011}r_{1010}})}, \\
y_8 = \frac{r_{0111}r_{1001}r_{1100}}{r_{1101}\delta}, \\
y_9 = \frac{r_{1101}(r_{0101}\delta-r_{0111}r_{1101})}{r_{1101}\delta-r_{0111}r_{1001}r_{1100}}.
\end{cases}
\tag{3A}
$$

Thus, the parameter set $\{y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8, y_9, \gamma\}$ on CDM 5 is identifiable.

□

Since CDMs $1-4$ are all nested in CDM 5, the transformed phylogenetic tensors of CDMs $1-4$ can be determined directly from that of CDM 5 by setting some parameters x_i to 1. Similarly, it is straightforward to prove that the equivalent sets of y_i parameters are identifiable for each of CDMs $1-4$. The numbers of degrees of freedom for the phylogenetic tensors of CDMs $1-5$ are 6, 7, 8, 9 and 10, respectively.

3A Proof of Theorem 2

Recall from Definition 5 that if all points in the intersection of the sets of possible realizable phylogenetic tensors correspond to non-generic parameters in at least one of the parameters spaces of the CDMs, then the CDMs are distinguishable. We provide a technical justification for this definition here.

Recall that CDMs $\mathcal{N}_1$ and $\mathcal{N}_2$ have sets of possible realizable phylogenetic tensors $\mathcal{P}_1$ and $\mathcal{P}_2$. Then $\mathcal{P}_1$ is defined by a system of polynomial equations (phylogenetic invariants) in the form $g_i = 0$ and inequalities (parameter constraints). Then suppose we have affine variety $\mathcal{V}_1 = \mathcal{V}_1(g_1, g_2, \ldots, g_s) = \{(a_1, a_2, \ldots, a_r) \in \mathbb{Q}^r | g_i(a_1, a_2, \ldots, a_r) = 0 \text{ for all } i \in \{1, 2, \ldots, s\}\}$.

Recall that Θ_1' corresponds to parameters in Θ_1 corresponding to realizable phylogenetic tensors in $\mathcal{P}_1 \cap \mathcal{P}_2$. Suppose Θ_1' corresponds to non-generic parameters in Θ_1. Suppose also that $\mathcal{V}_1'$ is the affine variety defined by the system of polynomials for $\mathcal{P}_1$ restricted to $\mathcal{P}_1 \cap \mathcal{P}_2$. Then $\mathcal{V}_1' = \mathcal{V}_1 \cap H_1 \cap H_2 \cap \ldots \cap H_w$, where $H_1, H_2, \ldots, H_w$ are affine varieties defined by hypersurfaces (one polynomial equation each). By Exercise 1.8 of Chapter 1 of Hartshorne (2013), every irreducible component of $\mathcal{V}_1'$ has dimension $\dim(\mathcal{V}_1) - w$. By Corollary 9 of Chapter 9, Section 4 of Cox et al. (1997), $\dim(\mathcal{V}_1')$ is the largest of the dimensions of its irreducible components. Thus, $\dim(\mathcal{V}_1') = \dim(\mathcal{V}_1) - w$. Thus, the set of points in $\mathcal{P}_1 \cap \mathcal{P}_2$ is "small" compared to the set of points in $\mathcal{P}_1$.

For a more in-depth proof of Theorem 2, we could consider the (reduced) Gröbner bases of the ideals representing the parameter spaces of the CDMs and show that each CDM has a unique Gröbner basis. The Gröbner basis for CDM 5 has already been computed in Section 2A. However, computation of the Gröbner bases is slow and some bases contain many generators. Instead, it is sufficient to consider only a few constraints for each parameter space that exist for some CDMs and not others, greatly simplifying the proof.

To prove distinguishability, we show that only non-generic (or possibly no) parameters of one CDM correspond to points in the intersection of possible realizable phylogenetic tensors for two CDMs. To do this, we could show that there are polynomial equations (phylogenetic invariants) involving the elements of the transformed phylogenetic tensor $(q_{0000}, q_{0001}, \ldots, q_{1111})$ that exist for one CDM, but not the other. However, noting that $\gamma = q_{0001} = q_{0010} = q_{0100} = q_{1000}$ for all our 4-taxon CDMs, it is easier to find polynomial equations involving the elements $r_{0000}, r_{0001}, \ldots, r_{1111}$ and δ and use these to establish distinguishability.

Theorem 2 *All pairs of* 4*-taxon leaf-labeled CDMs of Section 4.2 are distinguishable.*

Proof Suppose first that one CDM is nested in the other. With no loss of generality, assume CDM $\mathcal{N}_2$ with parameter space Θ_2 and set of possible realizable phylogenetic tensors $\mathcal{P}_2$ is nested in CDM $\mathcal{N}_1$ with parameter space Θ_1 and set of possible realizable phylogenetic tensors $\mathcal{P}_1$. Then Θ_2 can be obtained from Θ_1 by fixing some parameter(s). However, since Θ_2 corresponds to fixing some parameter(s) to be on the boundary of Θ_1 (setting edge lengths or epochs to zero) and is thus in the closure of Θ_1 but not Θ_1 itself, by Assumption 12, $\Theta_1 \cap \Theta_2 = \emptyset$. Thus, by the identifiability of all CDMs (see Proposition 1 for CDM 5), $\mathcal{P}_1 \cap \mathcal{P}_2 = \emptyset$. (Note that identifiability of $\mathcal{N}_1$ still holds if closure points are included in Θ_1.)

All that is left to prove is that any two CDMs that differ only in their leaf labelings are distinguishable. It then follows that two CDMs that differ in their leaf labelings and also their numbers of parameters are distinguishable. To see this, suppose $\mathcal{N}_1$ and $\mathcal{N}_2$ are distinguishable, with corresponding affine varieties $\mathcal{V}_1$ and $\mathcal{V}_2$ and corresponding sets of possible realizable phylogenetic tensors $\mathcal{P}_1$ and $\mathcal{P}_2$. Then $\mathcal{P}_1 \cap \mathcal{P}_2$ corresponds to a some lower dimensional subvariety of both $\mathcal{V}_1$ and $\mathcal{V}_2$. Then if $\mathcal{N}_1'$ is nested in $\mathcal{N}_1$, $\mathcal{P}_1' \cap \mathcal{P}_2$ must also correspond to some lower dimensional subvariety of $\mathcal{V}_2$.

The notation that follows is consistent with that of Section 2A. Recall that $y_i \in (0,1)$ for all $i \in \{1, 2, \ldots, 9\}$.

CDM 5

See S8.nb (text version S9.txt) on https://github.com/jonathanmitchell88/CDMsSI for proofs of the following claims.

For leaf labelings $(o,(a,(b,c)))$ and $(o,(a,(c,b)))$,

$$r_{0011}r_{1001}r_{1010} - r_{1011}^2 = 0,$$

while for the other leaf labelings

$$r_{0011}r_{1001}r_{1010} - r_{1011}^2 > 0.$$

Thus, if $\mathcal{P}_1$ is the set of realizable phylogenetic tensors corresponding to either leaf labeling $(o,(a,(b,c)))$ or $(o,(a,(c,b)))$ and $\mathcal{P}_2$ is the set of realizable phylogenetic tensors corresponding to a different leaf labeling, then $\mathcal{P}_1 \cap \mathcal{P}_2 = \emptyset$. (Note that the choice of taxon a as the sister taxon to the outgroup when the 4-taxon CDM is unrooted is arbitrary.)

Thus, we need only show that CDMs with leaf labelings $(o,(a,(b,c)))$ and $(o,(a,(c,b)))$ are distinguishable. Let $\mathcal{P}_1$ and $\mathcal{P}_3$ be the sets of realizable phylogenetic tensors corresponding to leaf labelings $(o,(a,(b,c)))$ and $(o,(a,(c,b)))$. We show that all points in $\mathcal{P}_1 \cap \mathcal{P}_3$ correspond to non-generic points in either Θ_1 or Θ_3. Letting y_i be the parameters corresponding to leaf labeling $(o,(a,(b,c)))$ and z_i corresponding to $(o,(a,(c,b)))$, we equate the equations for each element of the two phylogenetic tensors and solve for the z_i parameters,

$$\begin{cases} y_4y_5y_6y_8 = & z_4z_5z_6z_8, \\ y_9(1-y_8) + y_2y_3y_4y_6y_8 = & z_8(z_7(1-z_6) + z_2z_3z_5z_6), \\ y_8(y_7(1-y_6) + y_2y_3y_5y_6) = & z_9(1-z_8) + z_2z_3z_4z_6z_8, \\ y_2y_3y_4y_5y_6y_8 = & z_2z_3z_4z_5z_6z_8, \\ y_1y_2y_4y_8 = & z_1z_2z_5z_6, \\ y_1y_2y_5y_6 = & z_1z_2z_4z_8, \\ y_1y_2y_4y_5y_6y_8 = & z_1z_2z_4z_5z_6z_8, \\ y_1y_3y_6y_8 = & z_1z_3z_6z_8, \\ y_1y_2y_3y_4y_6y_8 = & z_1z_2z_3z_5z_6z_8, \\ y_1y_2y_3y_5y_6y_8 = & z_1z_2z_3z_4z_6z_8, \\ \begin{array}{r} y_1(y_4y_8(y_2y_7(1-y_6) + y_3y_5y_6) \\ +y_2y_5y_6y_9(1-y_8)) \end{array} = & \begin{array}{l} z_1(z_4z_8(z_2z_7(1-z_6) + z_3z_5z_6) \\ +z_2z_5z_6z_9(1-z_8)). \end{array} \\ y_1y_2y_3y_4y_5y_6y_8 = & z_1z_2z_3z_4z_5z_6z_8. \end{cases}$$

Solving this system of equations — see S10.m2 (output file S11.txt) and the expressions simplified in S8.nb (text version S9.txt) on https://github.com/jonathanmitchell88/CDMsSI — we obtain

$$z_1 z_2^2 z_3 z_4 z_6 z_8 \left(1 - z_6\right)\left(1 - z_8\right)\left(z_4 z_7 z_8 - z_5 z_9\right) = 0.$$

Since $z_i \in (0,1)$ for all $i \in \{1, 2, \ldots, 8\}$, we must have $z_4 z_7 z_8 - z_5 z_9 = 0$. These points are non-generic points in the parameter space. Thus, for CDM 5, any two CDMs with different leaf labelings are distinguishable.

CDM 4

The proof is identical to that of CDM 5, but with the addition of $y_9 = z_9 = 1$. Again, see S8.nb (text version S9.txt) and S10.m2 (output file S11.txt). We obtain

$$z_1 z_2 z_3 z_4 z_5 z_6 z_8 \left(1 - z_6\right)\left(1 - z_7 z_8\right) = 0,$$

which has no solutions since $z_i \in (0,1)$ for all $i \in \{1, 2, \ldots, 8\}$. Thus, for CDM 4, $\mathcal{P}_1 \cap \mathcal{P}_3 = \emptyset$ and any two CDMs with different leaf labelings are distinguishable.

CDM 3

See S8.nb (text version S9.txt) for proofs of the following claims.

For leaf labeling pairs $(o,(a,(b,c)))$ and $(o,(c,(b,a)))$, $(o,(a,(c,b)))$ and $(o,(b,(c,a)))$ and $(o,(b,(a,c)))$ and $(o,(c,(a,b)))$,

$$\begin{cases} r_{0101} r_{1010} < r_{0011} r_{1100}, r_{0110} r_{1001}, \\ r_{0110} r_{1001} < r_{0011} r_{1100}, r_{0101} r_{1010}, \\ r_{0011} r_{1100} < r_{0101} r_{1010}, r_{0110} r_{1001}, \end{cases}$$

respectively, where each equation corresponds to a leaf labeling pair. Thus, any CDM from one pair is distinguishable from a CDM from another pair.

All that is left is to prove that CDMs from an arbitrary pair, for example, $(o,(a,(b,c)))$ and $(o,(c,(b,a)))$, are distinguishable. For leaf labeling $(o,(a,(b,c)))$, but not $(o,(c,(b,a)))$,

$$r_{0011}r_{1001}r_{1010} - r_{1011}^2 = 0.$$

For leaf labeling $(o,(c,(b,a)))$, but not $(o,(a,(b,c)))$,

$$r_{0011}r_{1001}r_{1010} - r_{1011}^2 > 0.$$

Thus for CDM 3, any two CDMs with different leaf labelings are distinguishable.

CDM 2

See S8.nb (text version S9.txt) for proofs of the following claims.

The constraints for CDM 2 include those described above for CDM 3. Thus for CDM 2, any two CDMs with different leaf labelings are distinguishable.

CDM 1

See S8.nb (text version S9.txt) for proofs of the following claims.

For leaf labelings $(o,(a,(b,c)))$, $(o,(b,(a,c)))$ and $(o,(c,(a,b)))$,

$$\begin{cases} r_{0101}r_{1010} = r_{0110}r_{1001} < r_{0011}r_{1100}, \\ r_{0011}r_{1100} = r_{0110}r_{1001} < r_{0101}r_{1010}, \\ r_{0011}r_{1100} = r_{0101}r_{1010} < r_{0110}r_{1001}, \end{cases}$$

respectively. Thus, for CDM 1 any two CDMs with different leaf labelings are distinguishable.

□

4A Proof of Theorem 3

Theorem 3 (Distance on the topology of an N-taxon principal tree) *Let $\mathcal{T}$ be a principal tree, with outgroup o. Suppose $\mathcal{T}$ is given the rooted triple metrization. Then the distance*

$d_{\mathcal{T}}(x,y)$ between leaf taxa x and y is

$$d_{\mathcal{T}}(x,y) = \begin{cases} 0 & \text{if } x{=}y, \\ 2N-2 & \text{if } x \neq y \text{ and one of } x=o,\ y=o, \\ 2\left|R_{x,y}\right|+2 & \text{otherwise,} \end{cases}$$

where $R_{x,y}$ is the set of rooted 4-taxon principal trees displayed on $\mathcal{T}$ with outgroup o displaying both x and y, where x and y are non-sisters.

Proof Clearly, if $x = y$ then $d_{\mathcal{T}}(x,y) = 0$.

Next suppose $x \neq y$ and one of $x = o$, $y = o$. With no loss of generality, assume $y = o$. Then

$$d_{\mathcal{T}}(x,y) = d_{\mathcal{T}}(x,o) = d_{\mathcal{T}}(x,v) + d_{\mathcal{T}}(v,o),$$

where v is the most recent common ancestor (MRCA) of x and o. Since $y = o$, v must be the root of $\mathcal{T}$. Then from the rooted triple metrization, by the same arguments as Rhodes (2019),

$$d_{\mathcal{T}}(x,v) = d_{\mathcal{T}}(v,o) = N-1$$

and

$$d_{\mathcal{T}}(x,o) = 2N-2.$$

Finally, suppose $x \neq y$ and $x, y \neq o$. Again suppose that v is the MRCA of x and y. Then again by the same arguments as Rhodes (2019),

$$d_{\mathcal{T}}(x,y) = 2k-2,$$

where k is the number of leaf taxa descended from v.

For x and y to be non-sisters on a rooted 4-taxon principal tree displayed on $\mathcal{T}$ with outgroup o, we require the leaf taxon that is not x, y or o to be one of the $k-2$ leaf taxa descended from v that is not x or y. Thus,

$$\left|R_{x,y}\right| = k-2$$

and

$$d_{\mathcal{T}}(x,y) = 2\left|R_{x,y}\right| + 2.$$

$\square$

5A Inferring topologies of N-taxon principal trees

We prove that consistent inference of the topology of the N-taxon principal tree follows from consistent inference of the principal trees of the displayed 4-taxon CDMs. However, it is possible that a displayed 4-taxon CDM does not meet the assumptions of Section 3.2. Specifically, even if an N-taxon CDM meets the assumptions, some displayed 4-taxon CDMs may have sister convergence. By assuming that all convergence parameters of the N-taxon CDM are sufficiently "small", then all convergence parameters of the displayed 4-taxon CDMs, including those of sister convergence groups of the displayed 4-taxon CDMs are "small". Then all topologies of the displayed 4-taxon principal trees are inferred consistently by Algorithm 1.

To prove this result, we first prove a proposition similar to Proposition 1.2 of Haughton (1988). Proposition 1.2 states that if the generating model is among the set of candidate models, the probability that the model selected by the BIC is the generating model converges to 1. Our adaptation relaxes Proposition 1.2, such that none of the candidate models are the generating model, but some candidate models are sufficiently "close" to the generating model. That is, the generating parameter is a "small" perturbation from a point in the parameter space of a candidate model. We then use our proposition to prove that all topologies of the displayed 4-taxon principal trees are inferred consistently by Algorithm 1.

For the following proposition, $f(X,\phi) = \exp\left(X\phi - b(\phi)\right)$ is the density for a regular exponential family, m_1 and m_2 are the natural parameter spaces of two models, $\operatorname{int}\Theta$ is the interior of some topological space Θ, $\overline{m}_1$ and $\overline{m}_2$ are the Zariski closures of

m_1 and m_2, respectively and $E_\theta X_i = \nabla b(\theta)$ is the expected value of random variable X_i given generating parameter θ. The natural parameter space of the exponential family is as defined in Lehmann and Romano (2005), page 51. Note that for our multinomially distributed random variables, the natural parameter space is the set of realizable phylogenetic tensors. The function $g(\phi) = \nabla b(\theta)\phi - b(\phi)$ for $\phi \in \Theta$ attains its unique maximum at θ (Barndorff-Nielsen 1978).

Proposition 2A *Let m_1 and m_2 be two different models satisfying $m_1 \cap m_2 = \emptyset$. Then there exists some $\theta \in \operatorname{int}\Theta$, $\theta \notin \overline{m}_1, \theta \notin \overline{m}_2$, with a neighborhood $\mathfrak{N}$ of θ such that $\mathfrak{N} \cap m_1 = \emptyset$, $\mathfrak{N} \cap m_2 \neq \emptyset$ and*

$$\lim_{n\to\infty} P_\theta^n (\gamma(n,1) < \gamma(n,2)) = 1.$$

Proof The proof requires only a slight modification to the proof of Proposition 1.2 of Haughton (1988).

From Haughton (1988), since $\mathfrak{N} \cap m_1 = \emptyset$,

$$\sup_{\phi\in m_1\cap\Theta} \nabla b(\theta)\phi - b(\phi) + \epsilon \leq \nabla b(\theta)\theta - b(\theta) \tag{4A}$$

and asymptotically with probability 1,

$$\left|\sup_{\phi\in m_i\cap\Theta} (Y_n\phi - b(\phi)) - \sup_{\phi\in m_i\cap\Theta} \nabla b(\theta)\phi - b(\phi)\right| < \frac{\epsilon}{4}, \tag{5A}$$

where $\epsilon > 0$.

Although $\mathfrak{N} \cap m_2 \neq \emptyset$, $g(\phi)$ attains its maximum at θ and $\theta \notin \overline{m}_2$. Thus, we can choose $\widetilde{\epsilon} > 0$ such that

$$\sup_{\phi\in m_2\cap\Theta} \nabla b(\theta)\phi - b(\phi) + \widetilde{\epsilon} = \nabla b(\theta)\theta - b(\theta). \tag{6A}$$

We consider the two possible signs of the argument of the absolute value in Inequality (5A). If

$$\sup_{\phi\in m_i\cap\Theta} (Y_n\phi - b(\phi)) - \sup_{\phi\in m_i\cap\Theta} \nabla b(\theta)\phi - b(\phi) \geq 0,$$

then from Inequality 5A,

$$\sup_{\phi \in m_1 \cap \Theta} \left(Y_n \phi - b\left(\phi\right)\right) < \sup_{\phi \in m_1 \cap \Theta} \nabla b\left(\theta\right) \phi - b\left(\phi\right) + \frac{\epsilon}{4}.$$

Similarly, if

$$\sup_{\phi \in m_i \cap \Theta} \left(Y_n \phi - b\left(\phi\right)\right) - \sup_{\phi \in m_i \cap \Theta} \nabla b\left(\theta\right) \phi - b\left(\phi\right) < 0,$$

then

$$\begin{aligned} \sup_{\phi \in m_1 \cap \Theta} \left(Y_n \phi - b\left(\phi\right)\right) &< \sup_{\phi \in m_1 \cap \Theta} \nabla b\left(\theta\right) \phi - b\left(\phi\right) \\ &< \sup_{\phi \in m_1 \cap \Theta} \nabla b\left(\theta\right) \phi - b\left(\phi\right) + \frac{\epsilon}{4}. \end{aligned}$$

Thus, from Inequalities (4A) and (5A), asymptotically with probability 1,

$$\sup_{\phi \in m_1 \cap \Theta} \left(Y_n \phi - b\left(\phi\right)\right) < \sup_{\phi \in m_1 \cap \Theta} \nabla b\left(\theta\right) \phi - b\left(\phi\right) + \frac{\epsilon}{4} \leq \nabla b\left(\theta\right) \theta - b\left(\theta\right) - \frac{3\epsilon}{4}. \tag{7A}$$

By similar arguments, from Inequality (5A) and Equation (6A), asymptotically with probability 1,

$$\sup_{\phi \in m_2 \cap \Theta} \left(Y_n \phi - b\left(\phi\right)\right) > \sup_{\phi \in m_2 \cap \Theta} \nabla b\left(\theta\right) \phi - b\left(\phi\right) - \frac{\epsilon}{4} = \nabla b\left(\theta\right) \theta - b\left(\theta\right) - \widetilde{\epsilon} - \frac{\epsilon}{4}. \tag{8A}$$

By Inequalities (7A) and (8A),

$$\begin{aligned} \sup_{\phi \in m_1 \cap \Theta} \left(Y_n \phi - b\left(\phi\right)\right) &< \nabla b\left(\theta\right) \theta - b\left(\theta\right) - \frac{3\epsilon}{4} \\ &= \nabla b\left(\theta\right) \theta - b\left(\theta\right) - \widetilde{\epsilon} - \frac{\epsilon}{4} + \widetilde{\epsilon} - \frac{\epsilon}{2} \\ &< \sup_{\phi \in m_2 \cap \Theta} \left(Y_n \phi - b\left(\phi\right)\right) + \widetilde{\epsilon} - \frac{\epsilon}{2} \\ &= \sup_{\phi \in m_2 \cap \Theta} \left(Y_n \phi - b\left(\phi\right)\right) - \delta, \end{aligned}$$

where $\delta = \frac{\epsilon}{2} - \widetilde{\epsilon}$.

If it is possible to choose $\delta > 0$, then asymptotically with probability 1,

$$\sup_{\phi \in m_1 \cap \Theta} \left(Y_n \phi - b\left(\phi\right)\right) + \delta < \sup_{\phi \in m_2 \cap \Theta} \left(Y_n \phi - b\left(\phi\right)\right).$$

We are free to choose any $\theta \in \operatorname{int}\Theta$. Thus, we choose θ to be an arbitrarily small perturbation from some point in m_2. Then $\widetilde{\epsilon} > 0$ is arbitrarily small and $\delta > 0$. The remainder of the proof then follows from Haughton (1988).

$\square$

A convergence group on the generating N-taxon CDM may be a sister convergence group on some displayed 4-taxon CDMs and a non-sister convergence group on others. Thus, we must assume that all convergence parameters of the generating N-taxon CDM are "small" relative to the divergence parameters.

Next, we adapt Theorem 3 of Steel (1992) to prove that the N-taxon principal tree can be identified from the set of 4-taxon principal trees that include the outgroup.

Theorem 3A (Steel, 1992) *For a set of rooted triples R, $\langle R \rangle = \{T\}$ if and only if R is consistent with T, and for each internal edge e of T there is a rooted triple in R which distinguishes e.*

The consequence of Theorem 3A of Steel (1992) is that if all trees of a set of (binary) rooted 3-taxon trees R are displayed on a (binary) rooted N-taxon tree T and each internal edge of T is an internal edge of at least one tree in R, then T is the only N-taxon tree that displays all the 3-taxon trees of R. In other words, the N-taxon tree T can be identified from the set of 3-taxon trees R.

Steel (1992) note that an analogous theorem exists for unrooted quartets. Thus, the *unrooted* principal tree of the N-taxon CDM can be identified from the set of $\binom{N-1}{3}$ topologies of *unrooted* 4-taxon principal trees that include the outgroup displayed on the *unrooted* principal tree of the N-taxon CDM. The principal tree of the N-taxon CDM is then rooted by the outgroup.

5A.1 Proof of Theorem 4

Finally, from Proposition 2A and Theorem 3A adapted to unrooted quartets that include the outgroup, we can prove Theorem 4.

Theorem 4 *Suppose CDM $\mathcal{N}$ has topology of principal tree $\mathcal{T}$. Suppose the BIC is used for model selection in step 2 of Algorithm 1. Suppose $\widehat{\mathcal{T}}$ is the estimate of $\mathcal{T}$ inferred by Algorithm 1. Then there exists some constant $c > 0$ such that if the largest convergence parameter of $\mathcal{N}$ is less than c,*

$$\lim_{n\to\infty} \mathbb{P}\left(\widehat{\mathcal{T}} = \mathcal{T}\right) = 1.$$

Proof Suppose $\mathcal{N}$ has a displayed 4-taxon CDM $\mathcal{N}_4$ with topology of principal tree $\mathcal{T}_4 = (o,(a,(b,c)))$. Then from the proof of Theorem 2, for $\mathcal{N}_4$,

$$r_{0011}r_{1001}r_{1010} - r_{1011}^2 = 0,$$

while for some 4-taxon CDM with topology of principal tree $\mathcal{T}_4' \neq (o,(a,(b,c)))$,

$$r_{0011}r_{1001}r_{1010} - r_{1011}^2 > 0.$$

Suppose m_1 corresponds to the union of sets of possible realizable phylogenetic tensors for CDMs $1-5$ for the topology of principal tree $\mathcal{T}_4$. Suppose also that m_2 corresponds to the union of sets of possible realizable phylogenetic tensors for CDMs $1-5$ for any 4-taxon topology of principal tree that is not $\mathcal{T}_4$. Then $m_1 \cap m_2 = \emptyset$. Suppose $\theta \notin \overline{m}_1, \overline{m}_2$. Then if $c > 0$ is sufficiently small, since the functions for the phylogenetic tensor elements are analytic, the phylogenetic tensor corresponding to the CDM generating parameter is in a neighborhood that includes a subset of m_1, but none of m_2. By Proposition 2A, m_1 is selected by the BIC asymptotically with probability 1.

Next, we prove the claim that the set of inferred topologies of 4-taxon principal trees equals the set of topologies of the principal trees of the 4-taxon CDMs displayed on $\mathcal{N}$. Then from the adaptation of Theorem 3A to unrooted quartets, the topology of the principal tree of

$\mathcal{N}$ is the only topology that displays all inferred 4-taxon principal trees. Thus, any consistent supertree inference method used in step 3 of Algorithm 1 infers the topology of the principal tree of $\mathcal{N}$ consistently and the proof is complete.

All that is left to prove is the claim that the probability of the set of inferred 4-taxon principal trees equalling the set of topologies of principal trees of 4-taxon CDMs displayed on $\mathcal{N}$ converges to 1.

Suppose A_i is the event where the topology of the i^{th} 4-taxon principal tree is inferred incorrectly, given some arbitrary order. Then, by Proposition 2A, there exists some sample size n such that for $n' > n$, $\mathbb{P}(A_i) < \epsilon_i$ for some arbitrarily small $\epsilon_i > 0$. Then by Boole's inequality,

$$\mathbb{P}\left(\cup_{i=1}^{\binom{N-1}{3}} A_i\right) \leq \sum_{i=1}^{\binom{N-1}{3}} \mathbb{P}(A_i) < \sum_{i=1}^{\binom{N-1}{3}} \epsilon_i,$$

an arbitrarily small positive quantity. Thus, the set of topologies of the inferred 4-taxon principal trees of step 2 of Algorithm 1 equals the set of topologies of the principal trees of the 4-taxon CDMs displayed on $\mathcal{N}$ with probability converging to 1.

$\square$

6A Controlling overfitting the CDM

The criteria $x^{(a)}$ and $y^{(a)}$ of Algorithm 2 limit overfitting of convergence groups to the inferred CDM. Further control of overfitting is achieved with a multiple comparisons correction, favoring 4-taxon trees over non-tree 4-taxon CDMs. For a given 4-taxon set that includes the outgroup taxon, the model selection criterion values are first converted into weights, for example, AIC or BIC (Burnham and Anderson 2004). These weights are a "tree weight" determined from the AIC or BIC of the tree and "non-tree weights" determined from the AIC or BIC values of the other CDMs. Tree weights could then be multiplied by some positive constant $b \geq 1$ to achieve further control for overfitting. A multiple comparisons correction, such as the Holm-Bonferroni method (Holm 1979), could then be applied to the tree weights over all 4-taxon sets that include the outgroup taxon, as if the weights were p-values. If the tree is "rejected", then the non-tree CDM with the lowest AIC or BIC is selected.

7A Proof of Proposition 5

Proposition 5 *For convergence group $C = \{c_1, c_2\}$ on CDM $\mathcal{N}$, let $a \in c_1$ and $b \in c_2$. Let v be the MRCA node of a and b, X_v be the set of leaf taxa descending from v and $X_C = c_1 \cup c_2$. Then the expected proportion of converging quartets for $\{a, b\}$ is*

$$\frac{|X_v \setminus X_C|}{N-3} = \frac{|X_v| - |X_C|}{N-3},$$

where $|X_v|$ and $|X_C|$ are the cardinalities of sets X_v and X_C.

Proof To determine the expected proportions of converging quartets, suppose taxa a and b are converging. Then convergence between these taxa can only be inferred on 4-taxon CDMs with topology of principal tree $(o, (a, (b, c)))$ or $(o, (b, (a, c)))$, for some arbitrary taxon c. With no loss of generality, we assume that the topology of the principal tree of some 4-taxon CDM is $(o, (a, (b, c)))$. To determine the expected proportions, we must determine the number of 4-taxon CDMs displayed on $\mathcal{N}$, displaying both a and b where they appear as non-sisters.

We start with the rooted tree $(o, (a, b))$ and append taxon c and include a convergence group C. One edge corresponding to the convergence group C must be ancestral to a, while the other must be ancestral to b. Thus, for C to be a non-sister convergence group, the remaining taxon c must be placed on an edge directly descended from v, corresponding to a speciation event before the epoch C is in. Thus, c could be any of the $|X_v \setminus X_C| = |X_v| - |X_C|$ taxa out of the $N - 3$ possible taxa that are not o, a or b. $\square$

8A Proof of Proposition 7

Proposition 7 *An arbitrary pair of distinct convergence groups on CDM $\mathcal{N}$ share no pair of converging leaf taxa.*

Proof Suppose C_1 and C_2 are two distinct convergence groups on $\mathcal{N}$. By Assumption 5 of Section 3.2, there can be at most one convergence group in each epoch. Thus, C_1 is either in an epoch before or after C_2. With no loss of generality, we assume that C_1 is in an epoch before C_2.

In order to share at least one pair of converging taxa, C_2 must be nested in C_1. However, by Assumption 9 of Section 3.2, there can be no convergence groups nested in other convergence groups.

□

9A Proof of Proposition 8

We assume that the topology of the principal tree of $\mathcal{N}$ is known. However, we note that if it is not known, from Theorem 4 it can be inferred consistently.

Proposition 8 *The set of all convergence groups on CDM $\mathcal{N}$ can be identified from the set of displayed 4-taxon CDMs after suppressing sister convergence groups.*

Proof The set of displayed 4-taxon CDMs after suppressing sister convergence groups defines a matrix of proportions of converging quartets. However, in general the set of all convergence groups on $\mathcal{N}$ cannot be identified from the matrix (see Figure 5). Instead, we can identify a *set* of possible sets of convergence groups on $\mathcal{N}$ that correspond to the matrix of proportions of converging quartets. Since the set of displayed 4-taxon CDMs after suppressing sister convergence groups is assumed known, for the remainder of the proof we can restrict to this set of sets of convergence groups. We must then prove that we can identify the specific set of all convergence groups of $\mathcal{N}$.

If $\mathcal{N}$ is a tree, then the set of displayed 4-taxon CDMs after suppressing sister convergence groups is a set of trees. Thus, by Corollary 6, the matrix of proportions of converging quartets is the zero matrix. Alternatively, if $\mathcal{N}$ is not a tree, then $\mathcal{N}$ must have at least one non-sister convergence group. Call one such non-sister convergence group $C = \{c_1, c_2\}$, with v the most recent common ancestral node of c_1 and c_2. Then by Proposition 5, the expected proportion of converging quartets for $a \in c_1$ and $b \in c_2$ is $\frac{|X_v|-|X_C|}{N-3}$, where X_v is the set of all taxa descending from v and $|X_C| = |c_1| + |c_2|$. By the definition of non-sister convergence groups, $|X_v| - |X_C| > 0$. Thus, the matrix of converging quartets is not the zero matrix. Thus, if $\mathcal{N}$ is a tree, the set of convergence groups can be identified from the set of displayed 4-taxon CDMs after suppressing sister convergence groups via the matrix of converging quartets.

For the remainder of the proof, we can assume that $\mathcal{N}$ is not a tree. Then the set of non-sister convergence groups defines a set S of 4-taxon CDMs displayed on $\mathcal{N}$ with non-sister convergence groups after suppressing sister convergence groups — note that 4-taxon CDMs of S can have one or two non-sister convergence groups. Suppose similarly that S' is a set of 4-taxon CDMs defined by a set of non-sister convergence groups not on $\mathcal{N}$ but with the same matrix of proportions of converging quartets as the set of non-sister convergence groups on $\mathcal{N}$. We must prove that there exists some 4-taxon CDM in S that is not in S'. Then we can identify the set of convergence groups on $\mathcal{N}$ from the set of 4-taxon CDMs.

We prove that there is some 4-taxon CDM in S that is not in S'. We first consider an arbitrary 4-taxon CDM $\mathcal{N}_4$ in S. Consider arbitrary leaf taxon pair $\{a, b\}$, where $a \in c_1$ and $b \in c_2$. Furthermore, assume $c \in X_v \setminus X_C$. Then with no loss of generality, we can assume the topology of the principal tree of $\mathcal{N}_4$ is $(o, (b, (a, c)))$.

Suppose that $C' = \{c'_1, c'_2\}$ is one such non-sister convergence group that defines S', with c'_1, c'_2, v', X'_v and $X_{C'}$ as in Proposition 5. Now consider 4-taxon CDM $\mathcal{N}'_4$, defined by C' and on leaf taxon set $\{o, a, b, c\}$, with topology of principal tree $(o, (b, (a, c)))$. Since we require a non-sister convergence group on $\mathcal{N}'_4$ where a and b are both converging, we must have either $a \in c'_1$ and $b \in c'_2$ or $a \in c'_2$ and $b \in c'_1$. With no loss of generality, we assume that $a \in c'_1$ and $b \in c'_2$. Then $c_1 \subseteq c'_1$ or $c_1 \supset c'_1$. Similarly, $c_2 \subseteq c'_2$ or $c_2 \supset c'_2$. Both v and v' are the MRCA of a and b. Thus $v' = v$.

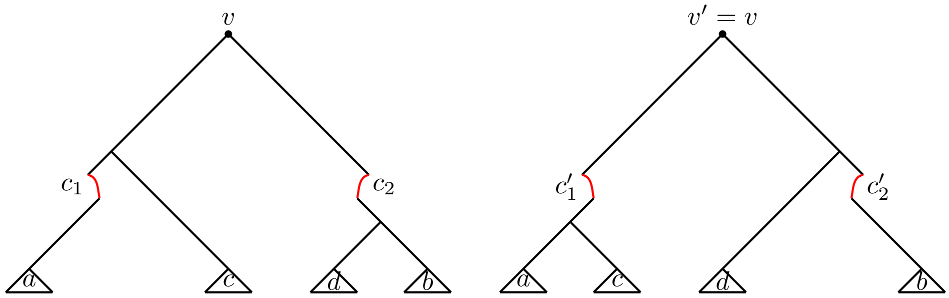
Now assume that $X_{C'} = X_C$. Then $c'_1 = c_1$ and $c'_2 = c_2$ and in turn, $C' = C$. Thus, S' is defined by a set of convergence groups that includes C and the 4-taxon CDM is in S'. Thus, we can assume that $X_{C'} \neq X_C$ and we cannot have both $c'_1 = c_1$ and $c'_2 = c_2$. However, since the matrices of proportions of converging quartets must be the same for the two sets of convergence groups, we must have

$$\frac{|X_v| - |X_C|}{N-3} = \frac{|X_{v'}| - |X_{C'}|}{N-3},$$

which simplifies to $|X_C| = |X_{C'}|$, since $v' = v$. Thus, either $c_1 \subset c_1'$ and $c_2 \supset c_2'$ or $c_1 \supset c_1'$ and $c_2 \subset c_2'$. With no loss of generality, we assume that $c_1 \subset c_1'$ and $c_2 \supset c_2'$.

Then there exists some choice of c such that $c \in c_1' \setminus c_1$. For $c_2 \supset c_2'$, there must similarly be some taxon $d \in c_2 \setminus c_2'$. Thus, we are assuming that $N \geq 5$ — the outgroup and taxa a, b, c and d. Then $a, c \in c_1'$ and $b \in c_2'$. Thus, before suppressing sister convergence groups to form $\mathcal{N}_4'$, C' must correspond with a sister convergence group on the 4-taxon CDM on leaf taxa $\{o, a, b, c\}$ — see Figure 1A for a graphical depiction of C and C'. Then any other choice of convergence group that defines S', say C'', must satisfy $c_1'' \supset c_1$ and the claim follows. Finally, since we have assumed $N \geq 5$, we must also consider $N = 4$. For $N = 4$, it is clear from the identifiability and distinguishability of all CDMs with no sister convergence that the claim holds.

□



(a) $C = \{c_1, c_2\}$ (b) $C' = \{c_1', c_2'\}$

Fig. 1A Convergence groups C and C'. Labels c_1, c_2, c_1' and c_2' indicate the set of leaf taxa below that edge. Triangles are displayed CDMs. Labels inside triangles indicate one of possibly many taxa on leaves of those displayed CDMs. There may be more displayed CDMs not drawn below v that are not below either C or C'

10A Proof of Theorem 9

Theorem 9 *Suppose CDM $\mathcal{N}$ has topology of principal tree $\mathcal{T}$ and convergence groups $\mathcal{G}$. Suppose for all l, $\alpha_l = \beta_l$. Suppose for convergence group $\mathcal{C}_i = \left\{c_{1,i}, c_{2,i}\right\}$ that if $a \in c_{1,i} \cup c_{2,i}$, then $a \notin c_{1,j} \cup c_{2,j}$ for any $j \neq i$. Suppose $\mathcal{T}$ is input into Algorithm 2, the BIC is used for model selection in step 2, there are no multiple comparisons corrections and one such tolerance criterion is $w = 1$. Suppose $\widehat{\mathcal{G}}$ is the estimate of $\mathcal{G}$ inferred by Algorithm 2. Then there exists some constant $c > 0$ such that if the largest convergence parameter of $\mathcal{N}$ is less than c,*

$$\lim_{n \to \infty} \mathbb{P}\left(\widehat{\mathcal{G}} = \mathcal{G}\right) = 1.$$

Proof We start by finding expressions for the transformed phylogenetic tensors for various 4-taxon CDMs with and without sister convergence. We prove that the CDMs with sister convergence are not distinguishable from the CDMs with the sister convergence groups suppressed. Thus, regardless of whether the 4-taxon CDMs have sister convergence groups or not, the non-sister convergence group is inferred consistently.

Since $\alpha_l = \beta_l$, $\gamma = 0$ and the transformed phylogenetic tensor for a 4-taxon CDM of Equation (1A) simplifies to

$$\widehat{P} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ r_{0011} \\ 0 \\ r_{0101} \\ r_{0110} \\ 0 \\ 0 \\ r_{1001} \\ r_{1010} \\ 0 \\ r_{1100} \\ 0 \\ 0 \\ r_{1111} \end{bmatrix}.$$

With no loss of generality, we assume the topology of the principal tree of an arbitrary 4-taxon CDM displayed on $\mathcal{N}$ is $(o\,(a,(b,c)))$. Then of the possible convergence groups on the 4-taxon CDM, the convergence group in the epoch closest to the root is the sister convergence group $C = \{\{a\}, \{b,c\}\}$. Thus, we first consider the distinguishability of two 4-taxon CDMs, one a tree and the other with this convergence group. For both CDMs we assume the tip epoch has epoch time 0. The first, which we call $\mathcal{N}_{4,1}$, is the tree $(o,(a,(b,c)))$. Since the tip epoch has epoch time 0, taxa b and c are identical. The second CDM, which we call $\mathcal{N}_{4,2}$, has a single convergence group, $C = \{\{a\}, \{b,c\}\}$, followed by a speciation event involving b and c. Again, since the tip epoch has epoch time 0, taxa b and c are identical. See Figure 2A for a graphical depiction of the two CDMs. Suppose $\mathcal{N}_{4,1}$ has parameters with no apostrophes and $\mathcal{N}_{4,2}$ has parameters with apostrophes.

For $\mathcal{N}_{4,1}$ (see Mathematica file S12.nb (text version S13.txt) on https://github.com/jonathanmitchell88/CDMsSI for a derivation),

$$\begin{cases} r_{0011} = & 1, \\ r_{0101} = & x_2 x_3, \\ r_{0110} = & x_2 x_3, \\ r_{1001} = & x_1 x_2, \\ r_{1010} = & x_1 x_2, \\ r_{1100} = & x_1 x_3, \\ r_{1111} = & x_1 x_3. \end{cases}$$

For $\mathcal{N}_{4,2}$ (see Mathematica file S12.nb (text version S13.txt) for a derivation),

$$\begin{cases} r_{0011} = & 1, \\ r_{0101} = & 1 - x_4' \left(1 - x_2' x_3'\right), \\ r_{0110} = & 1 - x_4' \left(1 - x_2' x_3'\right), \\ r_{1001} = & x_1' x_2' x_4', \\ r_{1010} = & x_1' x_2' x_4', \\ r_{1100} = & x_1' x_3' x_4', \\ r_{1111} = & x_1' x_3' x_4'. \end{cases}$$

For $\mathcal{N}_{4,1}$,

$$\begin{cases} x_1 = & \sqrt{\frac{r_{1001} r_{1100}}{r_{0101}}}, \\ x_2 = & \sqrt{\frac{r_{0101} r_{1001}}{r_{1100}}}, \\ x_3 = & \sqrt{\frac{r_{0101} r_{1100}}{r_{1001}}}, \\ r_{0101} = & r_{0110}, \\ r_{1001} = & r_{1010}, \\ r_{1100} = & r_{1111}. \end{cases}$$

Since $x_1, x_2, x_3 \in (0,1)$, for $\mathcal{N}_{4,1}$,

$$\begin{cases} r_{0101} = & r_{0110}, \\ r_{1001} = & r_{1010}, \\ r_{1100} = & r_{1111}, \\ r_{0101}r_{1001} < & r_{1100}, \\ r_{0101}r_{1100} < & r_{1001}, \\ r_{1001}r_{1100} < & r_{0101}. \end{cases}$$

Similarly, for $\mathcal{N}_{4,2}$,

$$\begin{cases} r_{0101} = & r_{0110}, \\ r_{1001} = & r_{1010}, \\ r_{1100} = & r_{1111}. \end{cases}$$

Since we are assuming that all convergence parameters of $\mathcal{N}$ are less than some constant $c > 0$, we can assume that $x'_4 = 1 - \epsilon$, where $\epsilon > 0$ is some small positive constant. Then for $\mathcal{N}_{4,2}$,

$$\begin{cases} r_{1100} - r_{0101}r_{1001} = & x'_1 x'_3 \left(1 - x_2'^2\right) + O\left(\epsilon\right), \\ r_{1001} - r_{0101}r_{1100} = & x'_1 x'_2 \left(1 - x_3'^2\right) + O\left(\epsilon\right), \\ r_{0101} - r_{1001}r_{1100} = & x'_2 x'_3 \left(1 - x_1'^2\right) + O\left(\epsilon\right). \end{cases}$$

Since $c > 0$ can be chosen, there exists some choice of $\epsilon > 0$ sufficiently small such that for $\mathcal{N}_{4,2}$,

$$\begin{cases} r_{0101} = & r_{0110}, \\ r_{1001} = & r_{1010}, \\ r_{1100} = & r_{1111}, \\ r_{0101}r_{1001} < & r_{1100}, \\ r_{0101}r_{1100} < & r_{1001}, \\ r_{1001}r_{1100} < & r_{0101}. \end{cases}$$

Thus, $\mathcal{N}_{4,1}$ and $\mathcal{N}_{4,2}$ are not distinguishable for this choice of $c > 0$. Thus, any 4-taxon CDM with $\alpha_l = \beta_l$ and this sister convergence group is not distinguishable from the CDM that results from suppressing the sister convergence. Thus, to determine the transformed phylogenetic tensor of any 4-taxon CDM with $\alpha_l = \beta_l$, we can assume there is no sister convergence in this epoch.

The next closest epoch to the root that could have a convergence group is the epoch just after taxa b and c have diverged. Thus, we compare the tree $(o, (a, (b, c)))$, which we call $\mathcal{N}_{4,3}$, and the CDM with topology of principal tree $(o, (a, (b, c)))$ and sister convergence group $\{\{b\}, \{c\}\}$ in the tip epoch, which we call $\mathcal{N}_{4,4}$. See Figure 3A for a graphical depiction of the two CDMs. Again, suppose $\mathcal{N}_{4,3}$ has parameters with no apostrophes and $\mathcal{N}_{4,4}$ has parameters with apostrophes.

For $\mathcal{N}_{4,3}$ (see Mathematica file S12.nb (text version S13.txt) for a derivation),

$$\begin{cases} r_{0011} = & x_4 x_5, \\ r_{0101} = & x_2 x_3 x_4, \\ r_{0110} = & x_2 x_3 x_5, \\ r_{1001} = & x_1 x_2 x_4, \\ r_{1010} = & x_1 x_2 x_5, \\ r_{1100} = & x_1 x_3, \\ r_{1111} = & x_1 x_3 x_4 x_5. \end{cases}$$

For $\mathcal{N}_{4,4}$ (see Mathematica file S12.nb (text version S13.txt) for a derivation),

$$\begin{cases} r_{0011} = & 1 - x'_6\left(1 - x'_4 x'_5\right), \\ r_{0101} = & x'_2 x'_3 x'_4 x'_6, \\ r_{0110} = & x'_2 x'_3 x'_5 x'_6, \\ r_{1001} = & x'_1 x'_2 x'_4 x'_6, \\ r_{1010} = & x'_1 x'_2 x'_5 x'_6, \\ r_{1100} = & x'_1 x'_3, \\ r_{1111} = & x'_1 x'_3\left(1 - x'_6\left(x'_4 x'_5\right)\right). \end{cases}$$

For $\mathcal{N}_{4,3}$,

$$\begin{cases} x_1 = & \sqrt{\frac{r_{1001} r_{1100}}{r_{0101}}}, \\ x_2 = & \sqrt{\frac{r_{0110} r_{1001}}{r_{0011} r_{1100}}}, \\ x_3 = & \sqrt{\frac{r_{0101} r_{1100}}{r_{1001}}}, \\ x_4 = & \sqrt{\frac{r_{0011} r_{0101}}{r_{0110}}}, \\ x_5 = & \sqrt{\frac{r_{0011} r_{0110}}{r_{0101}}}, \\ r_{0101} r_{1010} = & r_{0110} r_{1001}, \\ r_{0011} r_{1100} = & r_{1111}. \end{cases}$$

Since $x_1, x_2, x_3, x_4, x_5 \in (0,1)$, for $\mathcal{N}_{4,3}$,

$$\begin{cases} r_{0101}r_{1010} = & r_{0110}r_{1001}, \\ r_{0011}r_{1100} = & r_{1111}, \\ r_{0011}r_{0101} < & r_{0110}, \\ r_{0011}r_{0110} < & r_{0101}, \\ r_{0101}r_{1100} < & r_{1001}, \\ r_{0110}r_{1001} < & r_{0011}r_{1100}, \\ r_{1001}r_{1100} < & r_{0101}. \end{cases}$$

Similarly, for $\mathcal{N}_{4,4}$,

$$\begin{cases} r_{0101}r_{1010} = & r_{0110}r_{1001}, \\ r_{0011}r_{1100} = & r_{1111}. \end{cases}$$

Since we are assuming that all convergence parameters of $\mathcal{N}$ are less than some constant $c > 0$, we can assume that $x'_6 = 1 - \epsilon$, where $\epsilon > 0$ is some small positive constant. Then for $\mathcal{N}_{4,4}$,

$$\begin{cases} r_{0110} - r_{0011}r_{0101} = & x'_2x'_3x'_5\left(1 - x_4'^2\right) + O\left(\epsilon\right), \\ r_{0101} - r_{0011}r_{0110} = & x'_2x'_3x'_4\left(1 - x_5'^2\right) + O\left(\epsilon\right), \\ r_{1001} - r_{0101}r_{1100} = & x'_1x'_2x'_4\left(1 - x_3'^2\right) + O\left(\epsilon\right), \\ r_{0011}r_{1100} - r_{0110}r_{1001} = & x'_1x'_3x'_4x'_5\left(1 - x_2'^2\right) + O\left(\epsilon\right), \\ r_{0101} - r_{1001}r_{1100} = & x'_2x'_3x'_4\left(1 - x_1'^2\right) + O\left(\epsilon\right). \end{cases}$$

Since $c > 0$ can be chosen, there exists some choice of $\epsilon > 0$ sufficiently small such that for $\mathcal{N}_{4,4}$,

$$\begin{cases} r_{0101}r_{1010} = & r_{0110}r_{1001}, \\ r_{0011}r_{1100} = & r_{1111}, \\ r_{0011}r_{0101} < & r_{0110}, \\ r_{0011}r_{0110} < & r_{0101}, \\ r_{0101}r_{1100} < & r_{1001}, \\ r_{0110}r_{1001} < & r_{0011}r_{1100}, \\ r_{1001}r_{1100} < & r_{0101}. \end{cases}$$

Thus, $\mathcal{N}_{4,3}$ and $\mathcal{N}_{4,4}$ are not distinguishable for this choice of $c > 0$. Thus, any 4-taxon CDM with $\alpha_l = \beta_l$ and this sister convergence group is not distinguishable from the CDM that results from suppressing the sister convergence. Thus, to determine the transformed phylogenetic tensor of any 4-taxon CDM with $\alpha_l = \beta_l$, we can again assume there is no sister convergence in this epoch.

By the assumption that no leaf taxa belong to more than one convergence group, there can be no more than one convergence group on any arbitrary 4-taxon CDM displayed on $\mathcal{N}$. Thus, taking into consideration $\mathcal{N}_{4,1}$ and $\mathcal{N}_{4,2}$ not being distinguishable and $\mathcal{N}_{4,3}$ and $\mathcal{N}_{4,4}$ not being distinguishable, we can conclude that any arbitrary 4-taxon CDM displayed on $\mathcal{N}$ is not distinguishable from the 4-taxon CDM that results from suppressing any sister convergence group, which is one of CDM $1-3$ of Figure 2.

Next, we establish that CDM 3 is identifiable under these assumptions. For this CDM, which we call $\mathcal{N}_{4,5}$ (see Mathematica file S12.nb (text version S13.txt)),

$$\begin{cases} r_{0011} = & x_4x_5x_6x_7, \\ r_{0101} = & x_2x_3x_4x_6x_8, \\ r_{0110} = & x_7x_8\left(1 - x_6\left(1 - x_2x_3x_5\right)\right), \\ r_{1001} = & x_1x_2x_4, \\ r_{1010} = & x_1x_2x_5x_6x_7, \\ r_{1100} = & x_1x_3x_6x_8, \\ r_{1111} = & x_1x_4x_7x_8\left(x_2\left(1 - x_6\right) + x_3x_5x_6\right). \end{cases}$$

In terms of the set of parameters $\{y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8, y_9\}$ of Section 2A.1,

$$\begin{cases} r_{0011} = & y_4y_5y_6, \\ r_{0101} = & y_2y_3y_4y_6, \\ r_{0110} = & y_7\left(1 - y_6\right) + y_2y_3y_5y_6, \\ r_{1001} = & y_1y_2y_4, \\ r_{1010} = & y_1y_2y_5y_6, \\ r_{1100} = & y_1y_3y_6, \\ r_{1111} = & y_1\left(y_2y_4y_7\left(1 - y_6\right) + y_3y_5y_5y_6\right). \end{cases}$$

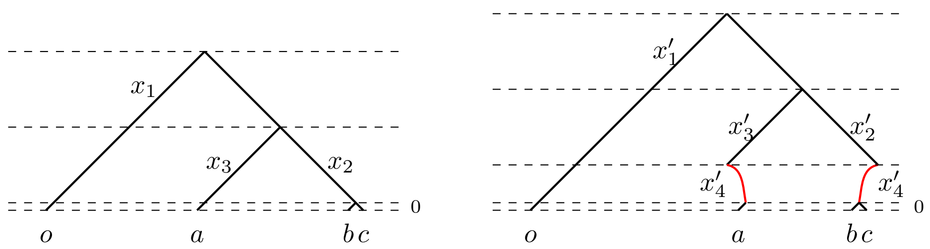
In S14.m2 (output file S15.txt) on https://github.com/jonathanmitchell88/CDMsSI, we see that the set of parameters $\{y_1, y_2, y_3, y_4, y_5, y_6, y_7\}$ is identifiable. It follows that CDMs 1 and 2 are also identifiable.

Thus, using similar arguments to those of the proof of Theorem 4, with probability converging to 1, step 2 of Algorithm 2 infers all the 4-taxon CDMs with the outgroup that are displayed on $\mathcal{N}$ after suppressing sister convergence groups.

If $\mathcal{N}$ is a tree, then $s = 0$ in step 4 of Algorithm 2, the algorithm terminates and the tree is returned. If $\mathcal{N}$ is not a tree, since $w = 1$, a potential convergence group on $\mathcal{N}$ is only considered if, for all pairs of converging taxa in the convergence group, the inferred 4-taxon

CDMs with that pair of taxa as non-sisters all have the pair converging. Thus, asymptotically with probability 1, only convergence groups on $\mathcal{N}$ can be on the inferred N-taxon CDM. If not all convergence groups of $\mathcal{N}$ have been included on the inferred CDM, then there are some elements of $\boldsymbol{O}$ that are non-zero corresponding to elements of $\boldsymbol{E}$ that are zero. These elements correspond to the pairs of converging taxa in convergence groups of $\mathcal{N}$ that are not yet on the inferred CDM. Including these convergence groups on the inferred CDM makes these elements of $\boldsymbol{E}$ equal to the corresponding elements of $\boldsymbol{O}$, decreasing the sum of squared differences. Once all convergence groups of $\mathcal{N}$ have been appended to the inferred CDM, $\boldsymbol{O} = \boldsymbol{E}$. Thus, no more convergence groups can be appended to the inferred CDM to decrease the sum of squared differences and the algorithm terminates.

$\square$



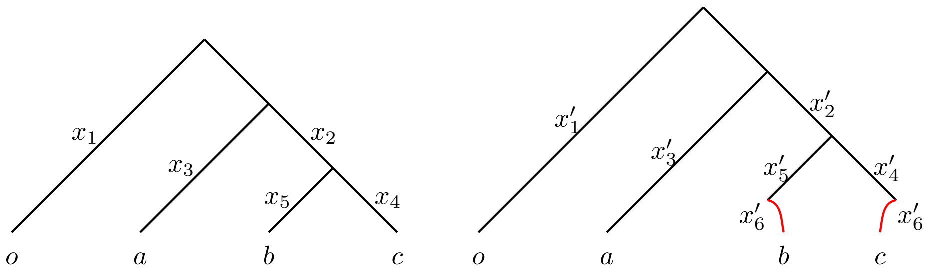
(a) $\mathcal{N}_{4,1}$ — x_1 represents the entire outgroup edge when unrooted.

(b) $\mathcal{N}_{4,2}$ — x'_1 represents the entire outgroup edge when unrooted.

Fig. 2A Two CDMs that are not distinguishable under the assumptions of Theorem 9

11A Inferring convergence group orders on N-taxon CDMs

The next algorithms infer partial orders on the convergence groups and determine whether or not there is a convergence group in the tip epoch. CDMs 4 and 5 have two convergence groups and thus provide power to determine convergence group orders. Whether or not there is a convergence group in the tip epoch can also be determined



(a) $\mathcal{N}_{4,3}$ (b) $\mathcal{N}_{4,4}$

Fig. 3A Two CDMs that are not distinguishable under the assumptions of Theorem 9

from the inferred 4-taxon CDMs. For example, CDM 2 and CDM 3 differ by CDM 2 having its convergence group in the tip epoch versus CDM 3 having its convergence group in the epoch before the tip epoch.

Suppose an inferred 4-taxon CDM has two non-sister convergence groups. One of the edges of the 4-taxon principal tree corresponds to a converging taxon in both convergence groups. The order of these convergence groups may not be determined by the matrix of edge partial orders from Algorithm 2. If the convergence group order is not determined, we determine which order is best supported by selecting CDMs from those with the appropriate convergence groups with a model selection procedure.

For convergence groups C_i and C_j, whose partial order has not been determined, we tally proportions of 4-taxon CDMs best supported by the two orders to obtain a matrix of "observed" convergence group order ratios. Convergence group orders are resolved in a stepwise fashion by minimizing the sum of squared differences between matrices of observed and "expected" partial convergence group order ratios. The matrix of inferred convergence group order ratios is updated after each convergence group order is inferred. We discard 4-taxon CDMs with convergence group orders not consistent with the matrix of inferred convergence group orders.

Suppose an arbitrary convergence group is $C_i = \{c_{1,i}, c_{2,i}\}$. On the N-taxon CDM, if $|c_{1,i}| > 1$ and/or $|c_{2,i}| > 1$ or C_i is in an epoch before another convergence group, then C_i cannot be in the tip epoch. For other convergence groups, whether they are in the tip epoch or not must be inferred.

For each 4-taxon CDM with a fixed leaf labeling with a possible convergence group in the tip epoch, we determine which CDM is best supported among the two CDMs, for example, CDM 2 versus CDM 3 or CDM 4 versus CDM 5. For C_i, we tally the 4-taxon CDMs displaying the given convergence group with and without the convergence group in the tip epoch.

If C_i corresponds to a greater proportion of 4-taxon CDMs with the convergence group in the tip epoch than any other convergence group and the proportion is greater than some cutoff, for example, half, then we infer that C_i is in the tip epoch. We retain only one possible CDM for each 4-taxon set after the convergence group order has been assigned and it has been determined which, if any, convergence group is in the tip epoch.

Note that some convergence group orders may still be undefined. Suppose two convergence groups do not have an order defined by the edge partial order of the principal tree or the orders of other convergence groups. Suppose both convergence groups are only ever present on 4-taxon CDMs where one convergence group is a sister convergence group. Then there will be no information to resolve the order of these two convergence groups. We leave these convergence group orders unresolved. Thus, we have a *partial* order on the convergence groups. Algorithms 1A and 2A for inferring convergence group orders and any convergence group in the tip epoch then follow.

We do not prove consistency of inference of the convergence group partial orders from Algorithm 1A. This is because Theorem 9 assumes that no leaf taxa belong to more than one convergence group. Thus, all 4-taxon CDMs displayed on $\mathcal{N}$ have at most one non-sister convergence group and there are no convergence group orders to infer. Furthermore, we do not prove consistency of inference of the convergence groups in the tip epoch.

Algorithm 1A Convergence group order inference

Input: N-taxon CDM $\widehat{\mathcal{N}}$ comprising N-taxon topology of principal tree $\widehat{\mathcal{T}}$ and list of convergence groups $\widehat{\mathcal{G}}$, as well as $\binom{N-1}{3} \times 27$ matrix of model selection criterion values $\boldsymbol{M}$ and matrix of partial edge orders $\boldsymbol{P}$.

1. Initialize empty list of inferred 4-taxon CDMs L_Q. Initialize $k \times k$ matrix of observed convergence group orders $\boldsymbol{O}$ as zero matrix, where k is length of list $\widehat{\mathcal{G}}$. Initialize $k \times k$ matrix $\boldsymbol{E}$ of expected convergence group orders as convergence group orders defined by $\boldsymbol{P}$, with $[\boldsymbol{E}]_{ij} = 1$ if convergence group i before j and 0 otherwise.
2. For each 4-taxon set that includes outgroup o, with model selection criterion, select CDM from those displayed on $\widehat{\mathcal{N}}$ and permitted by $\boldsymbol{E}$ and append to L_Q.
3. For all i, j, compute $[\boldsymbol{O}]_{ij}$ as proportion of inferred 4-taxon CDMs displaying convergence groups i and j, where i is before j.
4. Compute initial sum of squared differences between elements of $\boldsymbol{O}$ and $\boldsymbol{E}$, $s = \sum_{i=1}^{k} \sum_{j=1}^{k} \left([\boldsymbol{O}]_{ij} - [\boldsymbol{E}]_{ij} \right)^2$.
5. Assign new order between two convergence groups that minimizes s.
6. Update $\boldsymbol{E}$ and s to reflect newly inferred convergence group order. Suppose new order is convergence group x before y. Then all convergence groups above x are also above y and all convergence groups below y are also below x. If no pairs of convergence groups left to assign orders to, terminate algorithm.
7. Return to Step 5.

Output: N-taxon CDM $\widehat{\mathcal{N}}$ comprising N-taxon topology principal tree $\widehat{\mathcal{T}}$ and list of convergence groups $\widehat{\mathcal{G}}$, as well as $\binom{N-1}{3} \times 27$ matrix of model selection criterion values $\boldsymbol{M}$, matrix of partial edge orders $\boldsymbol{P}$ and matrix of expected convergence group orders $\boldsymbol{E}$.

However, if all inferred 4-taxon CDMs that include the outgroup are the 4-taxon CDMs displayed on the generating N-taxon CDM after suppressing sister convergence groups, then it is straightforward to prove that Algorithm 1A correctly infers all orders of convergence groups of the generating N-taxon CDM that can be determined from the displayed 4-taxon CDMs. Furthermore, it is also straightforward to prove that Algorithm 2A correctly infers which, if any, convergence group of the generating N-taxon CDM is in the tip epoch.

12A Proof of Proposition 10

Proposition 10 *All edge lengths of the principal tree of each of CDM $1-5$ are identifiable.*

Algorithm 2A Inference of convergence groups in tip epochs

Input: N-taxon CDM $\widehat{\mathcal{N}}$ comprising N-taxon topology principal tree $\widehat{\mathcal{T}}$ and list of convergence groups $\widehat{\mathcal{G}}$, as well as $\binom{N-1}{3} \times 27$ matrix of model selection criterion values $\boldsymbol{M}$, matrix of partial edge orders $\boldsymbol{P}$, matrix of expected convergence group orders $\boldsymbol{E}$ and tolerance $\tau \in [0,1]$.

1. Initialize empty list of inferred 4-taxon CDMs L_Q. Initialize vector $\boldsymbol{D}$ of length k of convergence groups in tip epoch as zero vector, where k is length of list $\widehat{\mathcal{G}}$.
2. For each 4-taxon set that includes outgroup o, select CDM from those displayed on $\widehat{\mathcal{N}}$ and permitted by $\boldsymbol{E}$ with model selection criterion and append to L_Q.
3. For all i, if convergence group $C_i = \{c_{1,i}, c_{2,i}\}$ satisfies $|c_{1,i}| = |c_{2,i}| = 1$ and is not before any other convergence group of $\widehat{\mathcal{N}}$, compute $[\boldsymbol{D}]_i$ as proportion of inferred 4-taxon CDMs with C_i in tip epoch.
4. If $\max_{i \in \{1,2,\ldots,k\}} [\boldsymbol{D}]_i = [\boldsymbol{D}]_j$ and $D_j > \tau$, set $[\boldsymbol{D}]_j = 1$.

Output: N-taxon CDM $\widehat{\mathcal{N}}$ comprising N-taxon topology principal tree $\widehat{\mathcal{T}}$ and list of convergence groups $\widehat{\mathcal{G}}$, as well as $\binom{N-1}{3} \times 27$ matrix of model selection criterion values $\boldsymbol{M}$, matrix of partial edge orders $\boldsymbol{P}$, matrix of expected convergence group orders $\boldsymbol{E}$ and vector of convergence groups in tip epoch $\boldsymbol{D}$.

Proof Using the parameterization of Section 2A.1, for CDM 5, with principal tree $(o,(a,(b,c)))$, the sums of edge lengths between leaf taxa are

$$\begin{cases} d_{o,a} = & l_1 + l_3 + l_6 + l_8 + l_9 + l_{11} = -\log\left(x_1 x_3 x_6 x_8 x_9 x_{11}\right) = -\log\left(y_1 y_3 y_6 y_8\right), \\ d_{o,b} = & l_1 + l_2 + l_5 + l_6 + l_7 = -\log\left(x_1 x_2 x_5 x_6 x_7\right) = -\log\left(y_1 y_2 y_5 y_6\right), \\ d_{o,c} = & l_1 + l_2 + l_4 + l_9 + l_{10} = -\log\left(x_1 x_2 x_4 x_9 x_{10}\right) = -\log\left(y_1 y_2 y_4 y_8\right), \\ d_{a,b} = & l_2 + l_3 + l_5 + 2l_6 + l_7 + l_8 + l_9 + l_{11} = -\log\left(x_2 x_3 x_5 x_6^2 x_7 x_8 x_9 x_{11}\right) \\ = & -\log\left(y_2 y_3 y_5 y_6^2 y_8\right), \\ d_{a,c} = & l_2 + l_3 + l_4 + l_6 + l_8 + 2l_9 + l_{10} + l_{11} = -\log\left(x_2 x_3 x_4 x_6 x_8 x_9^2 x_{10} x_{11}\right) \\ = & -\log\left(y_2 y_3 y_4 y_6 y_8^2\right), \\ d_{b,c} = & l_4 + l_5 + l_6 + l_7 + l_9 + l_{10} = -\log\left(x_4 x_5 x_6 x_7 x_9 x_{10}\right) = -\log\left(y_4 y_5 y_6 y_8\right). \end{cases}$$

From Equations (3A), the set $\{y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8, y_9\}$ is identifiable. Thus, the set $\{d_{o,a}, d_{o,b}, d_{o,c}, d_{a,b}, d_{a,c}, d_{b,c}\}$ is also identifiable for CDM 5. Solving for the lengths of the edges of the principal tree,

$$\begin{cases} l_o = & \frac{1}{2}\left(d_{o,a} + d_{o,b} - d_{a,b}\right), \\ l_a = & \frac{1}{2}\left(d_{o,a} - d_{o,b} + d_{a,b}\right), \\ l_b = & \frac{1}{2}\left(d_{a,b} - d_{a,c} + d_{b,c}\right), \\ l_c = & \frac{1}{2}\left(-d_{a,b} + d_{a,c} + d_{b,c}\right), \\ l_{bc} = & \frac{1}{2}\left(-d_{o,a} + d_{o,b} + d_{a,c} - d_{b,c}\right), \end{cases}$$

where l_o is the sum of divergence parameters along the two edges of the principal tree whose parent node is the root, l_a, l_b and l_c are the sums of divergence and possibly convergence parameters along the terminal edges whose descendent leaf taxa are a, b and c respectively and l_{bc} is the sum of divergence parameters along the edge whose descendent leaf taxa are b and c.

It follows that all edge lengths are also identifiable for CDMs $1-4$ since expressions for the sums of edge lengths are the same, except that some $y_i = 1$.

□

13A Proof of Proposition 11

Proposition 11 *All convergence parameters of each of CDM $2-5$ are identifiable.*

Proof On CDM 5, parameters $y_6 = x_6$ and $y_8 = x_9$ are identifiable. Thus, the convergence parameters $l_6 = a_6 + b_6 = -\log(y_6)$ and $l_9 = a_9 + b_9 = -\log(y_8)$ are identifiable. Thus, for all other CDMs with these convergence parameters, they are also identifiable. □

14A Proof of Proposition 12

Proposition 12 *The root parameter $\gamma = [\mathbf{\Pi}]_0 - [\mathbf{\Pi}]_1$, where $[\mathbf{\Pi}]_0$ and $[\mathbf{\Pi}]_1$ are the probabilities of states 0 and 1 at the root, respectively, is identifiable on each of CDM $1-5$.*

Proof From Equation (1A) for the phylogenetic tensor of CDM 5, $q_{0001} = q_{0010} = q_{0100} = q_{1000} = \gamma$. Thus, γ is identifiable for CDM 5. Since all other CDMs are nested in CDM 5 and none correspond to generic values of γ — instead they correspond to some generic values of x_i or y_i — γ is also identifiable for CDMs $1-4$. □

15A Proof of Theorem 13

Theorem 13 *Suppose CDM $\mathcal{N}$ has topology of principal tree $\mathcal{T}$, convergence groups $\mathcal{G}$, principal tree edge lengths $\boldsymbol{l}$, root parameter γ and convergence parameters $\boldsymbol{v}$. Suppose $\mathcal{T}$, $\mathcal{G}$, convergence group partial orders and tip epoch convergence groups of $\mathcal{N}$ are input into Algorithm 3. Suppose in step 4 of Algorithm 3 only 4-taxon sets for which 4-taxon CDMs displayed on $\mathcal{N}$ have no sister convergence are considered. Suppose that for each convergence group of $\mathcal{G}$ — say $C_a = \{c_{1,a}, c_{2,a}\}$ — there is at least one 4-taxon CDM displayed on $\mathcal{N}$ with no sister convergence where $x \in c_{1,a}$, $y \in c_{2,a}$ are non-sister leaf taxa on the displayed CDM. Suppose further that matrix $\boldsymbol{X}$ in step 6 of Algorithm 3 has rank $2N-3$. Suppose $\widehat{\boldsymbol{l}}$, $\widehat{\gamma}$ and $\widehat{\boldsymbol{v}}$ are the estimates of $\boldsymbol{l}$, γ and $\boldsymbol{v}$, respectively, inferred by Algorithm 3. Then for any $\epsilon{>}0$,*

$$\lim_{n\to\infty} \mathbb{P}\left(\left|\widehat{\boldsymbol{l}} - \boldsymbol{l}\right| > \epsilon\right) = 0, \qquad \lim_{n\to\infty} \mathbb{P}\left(|\widehat{\gamma} - \gamma| > \epsilon\right) = 0, \qquad \lim_{n\to\infty} \mathbb{P}\left(|\widehat{\boldsymbol{v}} - \boldsymbol{v}| > \epsilon\right) = 0,$$

where $\left|\widehat{\boldsymbol{l}} - \boldsymbol{l}\right|$ and $|\widehat{\boldsymbol{v}} - \boldsymbol{v}|$ involve l^1 norms.

Proof In step 4 of Algorithm 3, only 4-taxon sets that include the outgroup for which 4-taxon CDMs displayed on $\mathcal{N}$ have no sister convergence are considered. Thus, all such 4-taxon CDMs displayed on $\mathcal{N}$ are CDM $1-5$. Since some 4-taxon sets may not be considered, we cannot yet assume that all parameters are identifiable. However, for a given 4-taxon set that is considered, from the proof of Proposition 10, all sums of edge lengths between leaf taxa in the 4-taxon set are identifiable. From Propositions 11 and 12, all convergence parameters on the 4-taxon CDM displayed on $\mathcal{N}$ and the root parameter γ are also identifiable.

Thus, for the given 4-taxon set, the estimates of sums of edge lengths between taxa formed from the sums of maximum likelihood estimates of parameters converge in probability to the sums of edge lengths between taxa for $\mathcal{N}$. Likewise, the maximum likelihood estimates of the convergence parameters converge in probability to the convergence parameters on $\mathcal{N}$ and the maximum likelihood estimate of γ also converges in probability to γ. Thus, it follows that when averaging over all 4-taxon sets that are considered, the estimates of the sums of edge lengths between taxa converge in probability to the values for $\mathcal{N}$.

Now, since the matrix $\boldsymbol{X}$ has rank $2N-3$, $\boldsymbol{X}^T\boldsymbol{X}$ is invertible. It follows that $\widehat{\boldsymbol{l}}$ also converges in probability to $\boldsymbol{l}$ in step 7 of Algorithm 3. By assumption, for each convergence group of $\mathcal{G}$ there is at least one 4-taxon CDM displayed on $\mathcal{N}$ where two converging taxa of the convergence group are non-sister taxa and there is no sister convergence. Thus, each convergence parameter of $\widehat{\boldsymbol{v}}$ is estimated at least once. Thus, $\widehat{\boldsymbol{v}}$ converges in probability to $\boldsymbol{v}$. Finally, since γ is fixed across all 4-taxon CDMs displayed on $\mathcal{N}$, to be consistently estimated it only needs to be estimated for one 4-taxon CDM displayed on $\mathcal{N}$. Thus, $\widehat{\gamma}$ converges in probability to γ.

$\square$

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